

Question	Answer	Mark	Mark scheme	Additional guidance
11 (a)	2	M1	for a correct start, using one rule of indices, eg $3^n \times 3^{12} = 3^{14}$ or $3^n \times 3^{20} = 3^{22}$ or $3^{20} \div 3^8 = 3^{12}$ or $3^{14} \times 3^8 = 3^{22}$ or $\frac{3^{n+20}}{3^8} = 3^{14}$ or $3^{n-8} \times 3^{20} = 3^{14}$ or for forming an equation in n , eg $n + 20 - 8 = 14$ oe or $(n =) 14 + 8 - 20$	
		A1	cao SCB1 for an answer of 3^2 if M0 scored	Allow an answer of $3^n = 3^2$ An answer of 9 or $3^n = 9, n \neq 2$ on its own is to be awarded 0 marks
(b)	$16m^8$	M1	for an intention to find the cube root and square, eg $\sqrt[3]{64m^{12}}^2$ or $\sqrt[3]{(64m^{12})^2}$ or $(4m^4)^2$ or $\sqrt[3]{4096m^{24}}$ or for am^8 with $a \neq 16$ or $16m^b$ with $b \neq 8$	Do not condone missing brackets 16 or m^8 imply M1 Allow multiplication sign for M1
		A1	cao	Accept $a = 16, b = 8$